

Aziz Laadhar

+1 (617) 595-2802 | aziz_laadhar@seas.harvard.edu | linkedin.com/in/aziz-laadhar | github.com/azizlaadhar

EDUCATION

EPFL — M.Sc. Data Science, Minor in Financial Engineering Sep 2023 – Present
Coursework: Stochastic Calculus, Econometrics, Portfolio Optimization, Quantitative Risk

EPFL — B.Sc. Communication Systems Sep 2020 – Jul 2023
Teaching Assistant: Operating Systems, Computer Architecture, Computer Networks.

QUANTITATIVE RESEARCH EXPERIENCE

Sirius Energy SA Lugano, Switzerland
Quantitative Research Analyst — Power and Gas Trading Jul 2024 – Feb 2025

- Developed a functional autoregressive supply/demand-curve model to forecast day-ahead power markets from 25M+ bids on a 500-point price grid; estimated three-lag curve operators by solving 750k-variable second-order cone programs.
- Converted forecast curves into clearing-price forecasts using the desk's EUPHEMIA market-coupling replica; reduced OOS hourly MAE from EUR 9.2/MWh to EUR 6.7/MWh versus the desk's model, with 59.5% lower maximum absolute error.
- Reverse-engineered Italy's undocumented D+1 import ceiling using a two-stage random-forest hurdle model with 42 features; rolling-window validation: F1 0.90, conditional R^2 0.89 vs. 0.72 for OLS; deployed daily to traders.
- Built a counterfactual day-ahead market-clearing backtest that re-solved the market under alternative inputs to attribute zonal price and net-position errors to offer curves, block orders, and transfer capacities.
- Labeled the generation technology of 34% of 658k unlabeled power-market offers via a MILP over 743 hours; identified latent PV supply from residual offers that tracked solar generation and concentrated at the low-price end of the stack.

RESEARCH EXPERIENCE

Harvard University Cambridge, MA
Research Scholar — Advisors: Prof. Jia Liu, Prof. Na Li Feb 2026 – Present

- Built self-supervised representation-learning components for high-dimensional spatiotemporal data, compressing multi-channel video into low-dimensional latent trajectories for behavioral-state inference.
- Designed a quantitative video-evaluation benchmark spanning eight scoring families; stress-tested its scoring rules under measurement noise and edge cases using 366 automated tests.

SLB — Schlumberger-Doll Research Cambridge, MA
Machine Learning Researcher — Mathematical Physics and Modeling Jul 2025 – Feb 2026

- Built a physics-constrained neural model for 2D electromagnetic resistivity inversion, adding a learned forward-surrogate data-consistency penalty during training; reduced median reconstruction error by 14.4% versus a supervised CNN baseline.
- Developed structure-preserving losses and adaptive sampling for noisy inverse problems, reducing surrogate data-misfit residuals by 35%; stress-tested robustness under noise and distribution shift through systematic ablations.

EPFL Visual Intelligence for Transportation Laboratory Lausanne, Switzerland
Research Assistant — Advisor: Prof. Alexandre Alahi Sep 2022 – Jun 2023

- Trained a graph-transformer policy for the constrained Dial-a-Ride Problem by imitation on 50k Gurobi solutions, achieving a 2.67% cost gap to exact optimization on 100 held-out instances.
- Trained a feasibility-masked reinforcement-learning policy using PPO, reducing violated ride-time windows from 1.61 to 0.33 per instance at a 2.37% cost gap; vectorized the Transformer's per-user encoding, cutting inference from 3.13 to 1.55 s/instance.

SELECTED PROJECTS

Mercuria Campus Power Trading Workshop (EPFL) — Winning Team Apr 2024

- Priced electricity caps across four Australian NEM regions using Monte Carlo with a mean-reverting jump-diffusion model and two Poisson jump processes; estimated parameters from 3M five-minute regional price observations.

Optiver EPFL Hackathon — Top 3 Team Dec 2023

- Built regression-based Tesla price/volatility signals and iterated feature, model, and trading-rule selection against simulated PnL.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch), C++, SQL, R; Linux, Git.

Quantitative Methods: Time-series forecasting, state-space models, Kalman filtering, ARIMA, functional autoregression, Monte Carlo, MLE, constrained optimization, market microstructure, backtesting.

Machine Learning: Tree ensembles, deep learning, physics-informed ML, graph neural networks, self-supervised learning, deep RL.

Languages: French and Arabic (native); English (C2).